

Rethinking Robust Multivariate Time Series Anomaly Detection: A Hierarchical Spatio-Temporal Variational Perspective

Xiao Zhang¹, Shuqing Xu¹, Huashan Chen¹, Zekai Chen¹, Fuzhen Zhuang¹, Hui Xiong², *Fellow, IEEE*, and Dongxiao Yu¹

Abstract—The robust multivariate time series anomaly detection can facilitate intelligent decisions and timely maintenance in various kinds of monitor systems. However, the robustness is highly restricted by the stochasticity in multivariate time series, which is summarized as *temporal stochasticity* and *spatial stochasticity* specifically. In this paper, we explicitly model the temporal stochasticity variables and the latent graph relationship variables into a unified graphical framework, which can achieve better robustness to dynamicity from both the spatial and temporal perspective. First, within the spatial encoder, every connection exists or not is modeled as a binary stochastic variable, and the graph structure can be learnt automatically. Then, the temporal encoder would embed the highly structured time series into latent stochastic variables to capture both complex temporal dependencies and neighbors information. Moreover, we design a history-future combined anomaly score mechanism with both reconstruction decoder and forecasting decoder to improve the anomaly detection performance. By weighting the historical anomaly factor, the future anomaly factor, and the prediction error of current timestamp, the anomaly detection at current timestamp could be more sensitive to anomaly detection. Finally, extensive experiments on three publicly available anomaly detection datasets demonstrate our proposed method can achieve

the best performance in terms of recall and F1 compared with state-of-the-arts baselines.

Index Terms—Anomaly detection, variational autoencoder, multivariate time series, self-attention.

I. INTRODUCTION

VAST amount of multivariate time series data are generated to monitor systems in various real-world scenarios, such as kinds of sensory devices from smart IoT infrastructures [1], smart power grids [2], smart transportation, intelligent industrial systems [3], etc. Anomalies are quite common in multivariate time series data [4], accurate and efficient anomaly detection can help administrators make right decisions and timely maintenance, which has drawn researchers' wide interests in both academia and industries in recent years [1], [5], [6], [7], [8].

Based on the data observations, anomalies are usually considered as unexpected values deviating from normal patterns significantly. Thus, to learn robust representations of normal patterns from multivariate time series is crucial for anomaly detection. However, the robustness in anomaly detection is highly restricted by the stochasticity in multivariate time series, which is summarized as *temporal stochasticity* and *spatial stochasticity* specifically in our opinions. The *temporal stochasticity* means that noises commonly exist in time series, which are usually caused by various small and random factors [9], which is shown in Fig. 1(a). The purpose of modeling temporal stochasticity is to enable the model to learn robust representations of normal patterns, thereby better distinguishing between normal and anomalous points [6], [10], [11], [12]. Some variational auto-encoders (VAE) [10], [11], [13], [14], [15], [16], [17], [18] or generative adversarial networks (GANs) [19], [20] based models have been proposed to model the temporal stochasticity. The LSTM-VAE [11] uses the LSTM as the encoder and decoder network of VAE to learn robust latent representations. Omni-Anomaly [10] proposes a stochastic recurrent neural network to learn the temporal dependence between latent representations. MTAD-GAT [21] maps the connected temporal and spatial feature learned by GAT to the latent space using a VAE. MAD-GAN [19] and TadGAN [20] use the GAN to model the temporal stochasticity and show good robustness.

The *spatial stochasticity* modeling aims to select reasonable similarity sequences dynamically for every time series, thereby

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Xiao Zhang, Shuqing Xu, and Dongxiao Yu are with the School of Computer Science and Technology, Shandong University, Qingdao 266237, China (e-mail: xiaozhang@sdu.edu.cn; sqxu@mail.sdu.edu.cn; dxyu@sdu.edu.cn).

Huashan Chen is with the Institute of Information Engineering, Chinese Academy of Sciences, Beijing 100093, China (e-mail: chenhuashan@iie.ac.cn).

Zekai Chen is with Amazon, Seattle, WA 98109-5210 USA (e-mail: zekaic@amazon.com).

Fuzhen Zhuang is with the Institute of Artificial Intelligence, SKLSDE, School of Computer Science, Beihang University, Beijing 100191, China (e-mail: zhuangfuzhen@buaa.edu.cn).

Hui Xiong is with the Thrust of Artificial Intelligence, The Hong Kong University of Science and Technology, Guangzhou 511458, China, and also with the Department of Computer Science and Engineering, The Hong Kong University of Science and Technology, Hong Kong, China (e-mail: xionghui@ust.hk).

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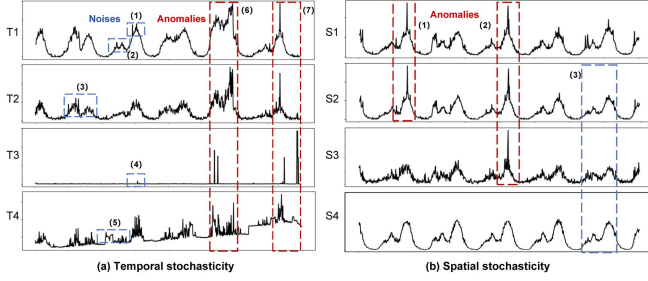


Fig. 1. An example of temporal stochasticity and spatial stochasticity. The blue blocks indicate noises and the red blocks indicate anomalies.

the learned graph relationships between sequences are dynamic with time changing [7]. As shown in Fig. 1(b), the similarity relationship is dynamically changing. Specifically, $S1$ and $S2$ are similar in region (1), while in region (2) and region (3), the similarities vary fast. Some previous works [5], [22] learn the graph structures by measuring similarities between embeddings and choosing top- K closest nodes as the spatial inter-relationships. This can easily lead to information loss on graph relationships or introduce inappropriate information to weaken the anomaly detection ability. Therefore, some methods [6], [7] considering spatial stochasticity have been proposed. For example in GRELEN [7], only latent graph relationship variables representing spatial stochasticity have been explicitly created, the latent variables representing temporal stochasticity have not been explicitly modeled in the framework. In summary, previous literature just utilize temporal latent variables to model the *temporal stochasticity* [10], [11], [12], or model the *spatial stochasticity* with a specific graph learning strategy [6], [7]. Hence in this work, we explicitly model the temporal stochasticity variables and the latent graph relationship variables into a unified graphical framework, which can achieve better robustness to dynamicity from both the spatial and temporal perspective.

Along this line, we propose a hierarchical spatio-temporal variational autoencoder framework to learn stochastic graph relations and stochastic latent representations simultaneously in an end-to-end manner. Within the spatial encoder, a directed graph structure can be learned by the graph learning strategy based on self-attention mechanism and Gumbel-Softmax sampling [1], [23]. Every connection exists or not would be modeled as a binary variable with Bernoulli prior assumption. The graph learning policy could automatically learn the hidden connections during training by deciding whether any specific node's information could flow to others to achieve the best performance. Then the temporal encoder embeds the highly structured time series into latent stochastic variables to capture both complex temporal dependencies and neighbor information. In details, the multi-head self-attention mechanism is utilized to obtain temporal feature representation for each time series and we use the graph attention networks (GAT) network to further aggregate the neighbor information for the temporal feature representation of each sequence according to the graph learned by the spatial encoder. Finally, since in real-world scenarios,

anomalies often occur in a continuous period of time, we design a history-future combined anomaly score mechanism with both reconstruction decoder and forecasting decoder. Specifically, we use the reconstruction decoder to reconstruct the historical time series data, and take the maximum reconstruction error in the history window as the historical anomaly factor at current timestamp. In the forecasting decoder, we adopt a multi-step prediction model to calculate the prediction values of current timestamp and the maximum prediction error of the future time window as the future anomaly factor. By weighting the historical anomaly factor, the future anomaly factor, and the prediction error of current timestamp, the anomaly detection at current timestamp could be more sensitive to improve the performance. Our main contributions are summarized as follows:

- We explicitly model the temporal stochasticity variables and the latent graph relationship variables into a unified graphical framework, which can achieve better robustness to dynamicity from both the spatial and temporal perspective.
- We propose an end-to-end hierarchical spatio-temporal variational autoencoder framework to address the *temporal stochasticity* and *spatial stochasticity* simultaneously. Within the spatial encoder, every connection exists or not is modeled as a binary stochastic variable, and the graph structure can be learnt automatically. Then the temporal encoder would embed the highly structured time series into latent stochastic variables to capture both complex temporal dependencies and neighbor information.
- We design a history-future combined anomaly score mechanism with both reconstruction decoder and forecasting decoder to improve the anomaly detection performance. By weighting the historical anomaly factor, the future anomaly factor, and the prediction error of current timestamp, the anomaly detection at current timestamp could be more sensitive for anomaly detection.
- We conduct extensive experiments on three real-world datasets to demonstrate the superiority of our framework over state-of-the-arts comparing with various multivariate time series anomaly detection baselines. We publicly publish our code on GitHub¹ for better reproducibility of our work.

II. RELATED WORK

Traditionally, time series anomaly detection algorithms can be divided into three learning types: supervised, unsupervised, and semi-supervised [24], [25]. Supervised methods [26] such as SVM [27], random forests [28] can achieve good performance due to the use of normal and anomalous labels. However labeling is not easy to get, which make supervised methods hardly practical. The unsupervised methods do not require labels and has good robustness against novel anomalies that have not appeared before. The KNN [29], K-means [30], DBStream [31] use the distance of observations to their nearest neighbors as anomaly score. In the COPOD [32], DSPOT [33], anomalies

¹<https://anonymous.4open.science/r/AD-HSTVAE-11C7>

TABLE I
CHARACTERISTIC SUMMARY OF THE BASELINES AND OUR HSTVAE

	Recon.	Fore.	Graph	Tem_stoch.	Spa_stoch.
PCA	✓	×	×	×	×
KNN	×	×	×	×	×
LSTM-VAE	✓	×	×	✓	×
OmniAnomaly	✓	×	×	✓	×
MAD-GAN	✓	×	×	✓	×
GRELEN	✓	×	✓	✓	✓
GDN	×	✓	✓	×	×
GTA	×	✓	✓	×	×
DVGCRN	✓	✓	✓	✓	✓
MTAD-GAT	✓	✓	✓	✓	✓
LSTM-AD	×	✓	×	×	×
HealthESN	×	✓	×	×	×
Telemanom	×	✓	×	×	×
HSTVAE (Ours)	✓	✓	✓	✓	✓

are considered extreme values in the distribution. The PCA [34] projects input to a lower-dimensional hyperplane and identifies data significantly deviating from this plane as anomalies. The semi-supervised methods use normal training data to guide the model to learn the normal patterns of the data, then distinguishing the data that deviates from the normal pattern in the testing set as anomalies. They usually consider the following aspects: temporal dependence, relationships between sequences, temporal stochasticity, and spatial stochasticity.

1) *Modeling Temporal Dependence*: A majority of works model the temporal dependencies of MTS through RNN network or attention mechanism. The GTA [1] models temporal dependency using a novel multi-branch attention mechanism. OmniAnomaly [10] proposes a stochastic recurrent neural network which can model the time dependencies between latent representations. LSTM-AD [35] designs stacked LSTM networks to learn the normal patterns, and use the predicted errors to judge anomalies. HealthESN [36] proposes a method based on the classical echo state network (ESN) and an adaptive detection threshold generation method based on information theory analysis. Telemanom [8] applies LSTM to anomaly detection in spacecraft telemetry and proposes a novel dynamic thresholding method.

2) *Modeling Graph Relationships Between Sequences*: MTAD-GAT [21] models the temporal and spatial dependencies simultaneously by combining a feature-oriented GAT and a time-oriented GAT. The GDN [22] learns the graph relationships between sequences by choosing the top- k neighbors, and uses a GAT to aggregate the selected neighbor feature. To overcome the limitations of top- k method, the GTA [1] proposes a graph learning policy based on the Gumbel-Softmax Sampling trick, which can learn the global bi-directed graph structure involving all sensors.

3) *Modeling Temporal Stochasticity and Spatial Stochasticity*: Many works utilize probability generation models to model temporal stochasticity. The LSTM-VAE [11], OmniAnomaly [10] utilize VAE to learn robust latent representations and capture the normal patterns of MTS. MAD-GAN [19], TadGAN [20] use the adversarial generative networks to model the temporal stochasticity. Some recent works also consider spatial stochasticity. DVGCRN [6] learns an embedding that follows a Gaussian distribution for each sequence to model the stochasticity inter

TABLE II
NOTATIONS USED IN THIS WORK

Indices	
N	The number of dimensions of a multivariate time series
w	Number of timestamps in the history sliding window
w'	Number of steps for multi-step prediction
x_i^t	Data of sequence i at timestamp t
$\hat{x}_i^t$	The reconstruction output of sequence i at timestamp t
y_i^t	The forecasting output of sequence i at timestamp t
d	Dimension of vector after embedding
d_g	Dimension of high-level extracted temporal features
d_z	The dimension of latent representation variable z_i
z_i	The latent representation for the sequence i
$\pi_{i,j}$	Probability parameters for the edge from sequence i to j
$e_{i,j}$	The latent variable for the edge from sequence i to j
ϕ_e	Parameters of spatial encoder
ϕ_z	Parameters of temporal encoder
θ_r	Parameters of reconstruction decoder
θ_f	Parameters of forecasting decoder

different sequences. GRELEN [7] proposes a self-attention-based graph relationship learning strategy and assumes the latent graph relationship variable obeys multinomial distribution. MTAD-GAT [21] combines the output of feature-oriented GAT and time-oriented GAT, and then maps the global features to a latent variable.

The comparison details are summarized in Table I. In conclusion, different from the existing works, our proposed HSTVAE can explicitly model the temporal stochasticity variables and the latent graph relationship variables into a unified graphical framework, which can achieve better robustness to dynamicity and achieve better anomaly detection performance.

III. PROBLEM STATEMENT

For the classical multivariate time series anomaly detection problem, we define $\mathbf{X} \in R^{N \times w}$ as the input of the multivariate time series data, where N denotes the dimensions of a multivariate time series, usually referring to the number of sensors within an entity, and w is the length of sliding time window. Let $x_i \in R^w$ denote the time series data at any sequence i , and $x^t \in R^N$ denote the data at any timestamp t . In addition, some notations are summarized in Table II.

Definition 1. Latent time series representation: To model temporal stochasticity, we assume that the input data of time series can be represented by a latent time series representation z that follows a distribution, i.e., $z \sim D_z(\phi_z)$, where D_z is the distribution that z follows, usually a Gaussian distribution, ϕ_z is the parameter of this distribution.

Definition 2. Latent graph relationship variable: To model the spatial stochasticity, we assume that the similarity relationship between sequences i and j can be represented by a latent graph relationship variable $e_{i,j}$ that follows a distribution, i.e., $e_{i,j} \sim D_e(\phi_e)$, where D_e is the distribution $e_{i,j}$ follows, usually can be a multinomial distribution, ϕ_e is the parameter of this distribution.

A semi-supervised model learns normal patterns on the D_{train} containing only normal data, and determines whether an anomaly occurred at time t on the D_{test} . For classical reconstruction or prediction based methods, when training, the

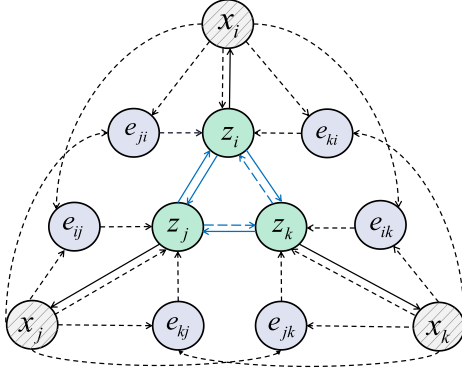


Fig. 2. Graphical model representations. The black dashed lines mean the inference model, the black solid lines mean the generative model. In addition, the blue solid arrow from latent representation z_i to z_j means there is an edge from node i to node j if the latent graph relationships variable $e_{ij} = 1$. And the blue dotted arrow from z_j to z_k means there is no edge from node j to node k if $e_{jk} = 0$.

goal is to minimize the following objective functions:

$$\mathcal{L} = \mathbb{E}_{X \sim D_{train}} [\|F(X; \theta, \phi_z, \phi_e) - \tilde{y}\|_2^2], \quad (1)$$

where θ is the model parameter, ϕ_z and ϕ_e are distribution parameter, $F(\cdot)$ represent the reconstruction or prediction network, $\tilde{y}$ represents the reconstructed or predicted groundtruth at the corresponding moment.

IV. MODEL ARCHITECTURE

A. Model Overview

In order to solve a common problem in the field of multivariate time series anomaly detection that the temporal fluctuations and relationship fluctuations can easily lead to misjudgment, we propose an end-to-end hierarchical spatio-temporal variational autoencoder framework called HSTVAE, which can simultaneously model the temporal stochasticity and spatial stochasticity of multivariate time series. As shown in Fig. 3, the overall framework consists of four modules. The spatial encoder uses the graph learning policy to discover the latent graph relationships between sequences, while the temporal encoder applies the multi-head self-attention mechanism and GAT network to capture the latent time series representation that considers both time dependence and neighbor information. The reconstruction decoder uses the multi-head self-attention network to model the temporal dependence in latent time series representations, and realizes the reconstruction of the input sequence. The prediction decoder uses the encode-decoder network to make every predicted timestamp in the decoder attend over all timestamps in the latent time series representations of input sequence and achieves accurate next-step prediction. In the back propagation process, the latent time series representation are updated reversely from two views of reconstruction and forecasting loss, which is helpful to learn more robust latent time series representation. As for the relationship between the above four modules, since different sequences might be highly correlated or similar and considering the neighbor information is meaningful,

we use the temporal encoder combining the graph relationship learned from the spatial encoder to learn richer latent time series representation z_i for each time series x_i . Then the reconstruction decoder constructs the historical sequence and the forecasting decoder predicts the future sequence according to the latent time series representations z_i and promote the learning of a more robust latent time series representations in the back propagation process.

It is worth noting that our model includes two different types of latent random variables: the edge-based latent graph relationships variable $e_{i,j}$ and sequence-based latent time series representation z_i . And the dependency of the two latent variables is shown in the Fig. 2. In Fig. 2, each node represents a random variable, the node with shadow represents the observed data, and the arrow represents the dependency between variables.

In the traditional VAE framework, the observation x is encoded as a latent variable z , and then the decoder reconstructs x from the z distribution. However, we can't compute the posterior distribution $p(z|x)$ analytically, thus the variational inference is proposed. The variational inference approximate the $p(z|x)$ using a variational distribution $q(z|x; \phi)$, whose parameters can be learned through an encoder network. We want to make $q(z|x; \phi)$ close to the posterior $p(z|x)$ and use the Kullback-Leibler (KL) divergence to measure the closeness of the two distributions. We can't minimize the KL divergence exactly, so we achieve this goal equivalently by maximizing ELBO. Actually, in many works [6], [10], [21] about variational inference, the evidence lower bound (ELBO) [37] function is the objective function which needs to be optimized, defined as

$$ELBO = \mathbb{E}_{q(z|x)} [\log p(x|z)] - D_{KL}[(q(z|x)||p(z))], \quad (2)$$

where the first term is the reconstruction loss and the second term is the Kullback-Leibler divergence loss.

In our HSTVAE framework, the ELBO of the i th sequence can be represented as

$$\begin{aligned} L(x_i, \phi_e, \phi_z, \theta_r, \theta_f) = & \mathbb{E}_{q_{\phi_z}(z_i|e_{:,i}, X)} [\log p_{\theta_r}(x_i|z_i, e_{:,i})] \\ & + \mathbb{E}_{q_{\phi_z}(z_i|e_{:,i}, X)} [\log p_{\theta_f}(y_i|z_i, e_{:,i})] \\ & - D_{KL}[q_{\phi_z}(z_i|e_{:,i}, X)||p(z_i)] \\ & - D_{KL}[q_{\phi_e}(e_{:,i}|X)||p(e_{:,i})]. \end{aligned} \quad (3)$$

Where the first term is the reconstruction loss which represents the ability of the reconstruction decoder to reconstruct the original input, and the second term is the forecasting loss which represents the prediction accuracy of the forecasting decoder. And the latter two terms represent the regularization terms that use the KL divergence to control the variation distribution and the prior distribution to be the same as possible. We use the variational distribution $q_{\phi_e}(e_{:,i}|X)$ to approximate the posterior distribution $p(e_{:,i}|X)$ for which does not have an analytic form. We assume $q_{\phi_e}(e_{:,i}|X)$ follows Bernoulli distribution and model the $q_{\phi_e}(e_{:,i}|X)$ using the spatial encoder. Similarly, we assume the variational distribution $q_{\phi_z}(z_i|e_{:,i}, X)$ follows Gaussian distribution and model the $q_{\phi_z}(z_i|e_{:,i}, X)$ using the temporal

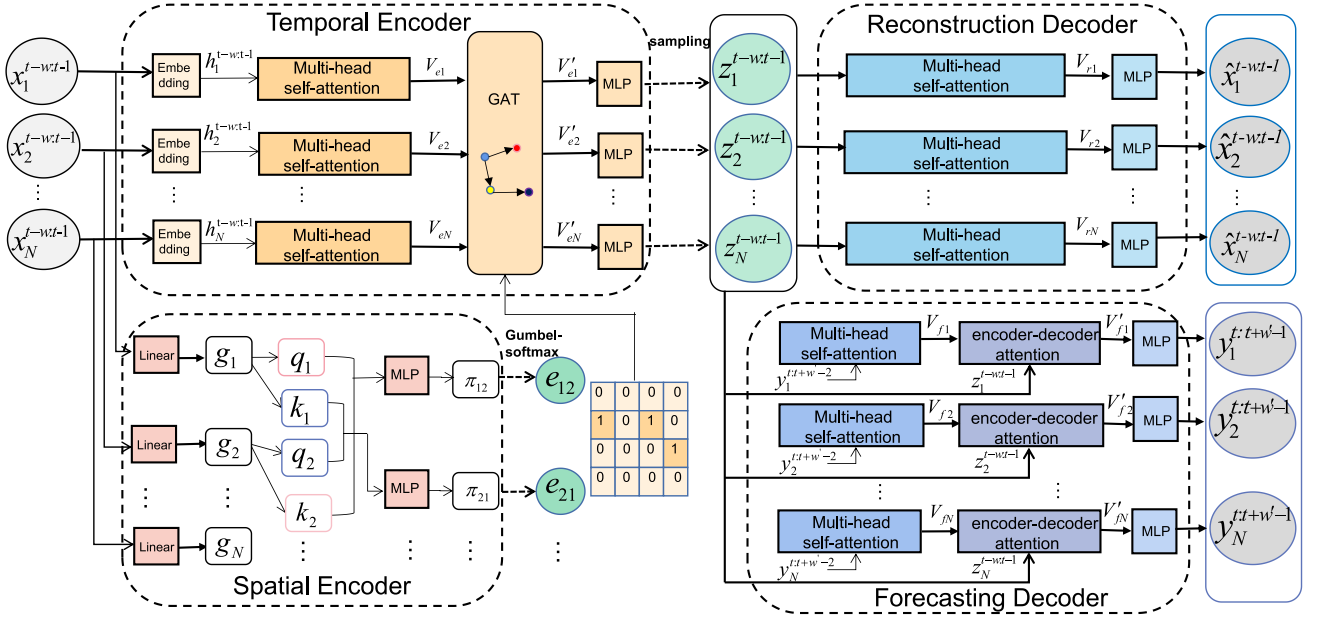


Fig. 3. The framework of the proposed HSTVAE which consists of a spatial encoder, a temporal encoder, a reconstruction decoder and a forecasting decoder. The four modules are related to each other through a latent time series representation and a latent graph relationships variable.

encoder. In the generative model, the $p_{\theta_r}(x_i | z_i, e_{:,i})$ is modeled by the reconstruction decoder and the $p_{\theta_f}(y_i | z_i, e_{:,i})$ is modeled by the forecasting decoder. The detailed derivation of (3) is shown as following.

$$\begin{aligned}
 \log p(X) &= \iint_{Z,E} q(Z, E | X) \log p(X) dZ dE \\
 &= \mathbb{E}_{q(Z,E|X)} \left[\log \frac{p(X, Z, E)}{p(Z, E | X)} \right] \\
 &\geq \mathbb{E}_{q(Z,E|X)} \left[\log \frac{p(Z, E) p(X | Z, E)}{q(Z, E | X)} \right] \\
 &= \iint_{Z,E} q(Z|E, X) q(E|X) \left[\log \frac{p(Z|E)}{q(Z|E, X)} \right] dZ dE \\
 &\quad + \iint_{Z,E} q(Z|E, X) q(E|X) \left[\log \frac{p(E)}{q(E|X)} \right] dZ dE \\
 &\quad + \iint_{Z,E} q(Z|E, X) q(E|X) [\log p(X|Z, E)] dZ dE \\
 &= \int_Z q(Z | E, X) \log p(X | Z, E) dZ \\
 &\quad + \int_Z q(Z | E, X) \log \frac{p(Z | E)}{q(Z | E, X)} dZ \\
 &\quad + \int_E q(E | X) \log \frac{p(E)}{q(E | X)} dE \\
 &= \mathbb{E}_{q(Z|E,X)} [\log p(X | Z, E)] \\
 &\quad - D_{KL}[q(Z | E, X) \| p(Z)] \\
 &\quad - D_{KL}[q(E | X) \| p(E)] = ELBO.
 \end{aligned}$$

B. Spatial Encoder Network

To model the stochastic inter-sequences graph relationships of multivariate time series, we propose the spatial encoder. Given two time sequences x_i and x_j , we use a latent graph relationships representation variable $e_{i,j}$ to represent the directed relationship from sequences x_i to x_j and assume the $q_{\phi_e}(e_{i,j} | x_i, x_j)$ follow a Bernoulli distribution. We use the spatial encoder to produce the parameter $\pi_{i,j}$ of the Bernoulli distribution. The details of the variational distribution are shown as follows:

$$q_{\phi_e}(e_{i,j} | x_i, x_j) = \text{Ber}(\pi_{i,j}(x_i, x_j)). \quad (4)$$

The $\pi_{i,j}$ is the probability of existing edge from node i to node j . We use the self-attention network as the spatial encoder, which can capture the similarity between sequences. First, we use a linear network to extract temporal feature $g_i \in R^{d_g}$ from each raw time series, that is $g_i = x_i W_g$, where $W_g \in R^{w \times d_g}$ is the learnable parameters. Then the extracted temporal feature g_i are mapped by learnable parameters matrices W_q and W_k to obtain q_i, k_i in different subspaces

$$q_i = g_i W_q, \quad k_i = g_i W_k. \quad (5)$$

Put the vector q_i and k_j into a weighted attention network to get a vector. Then the vector passes through an MLP network to get the probability variable $\pi_{i,j}$ whose value is between 0 to 1.

$$\pi_{i,j} = \text{MLP}(W_s [q_i, k_j]), \quad (6)$$

where W_s is the learnable parameters matrices.

Sampling of latent graph relationships: The Gumbel-Softmax distribution [1], [23] can be adapted to obtain a continuous approximation of Bernoulli random variables.

$$e_{i,j} = \frac{\exp((\log \pi_{i,j}^1 + g_{i,j}^1) / \tau)}{\sum_{v \in \{0,1\}} \exp((\log \pi_{i,j}^v + g_{i,j}^v) / \tau)}, \quad (7)$$

where $\pi_{i,j}^1 = \pi_{i,j}$, $\pi_{i,j}^0 = 1 - \pi_{i,j}$, $g_{i,j}^0, g_{i,j}^1$ are i.i.d samples drawn from a standard Gumbel distribution by drawing $u \sim \text{Uniform}(0, 1)$ and computing $g = -\log(-\log u)$. The temperatures τ controls the smoothness of Gumbel-Softmax distribution. When τ approaches 0, the probability of getting samples $e_{i,j}$ that are 0 or 1 approaches 1.

C. Temporal Encoder Network

To model the stochastic temporal dependencies of multivariate time series, we propose the temporal encoder. We assume the $q_{\phi_z}(z_i | e_{:,i}, X)$ follows a Gaussian distribution. Given an input time sequence x_i , the temporal encoder can produce the parameters μ_{z_i} and σ_{z_i} of the Gaussian distribution over the latent time series representation z_i . For the i th sequence, the details of the encoding process and variational distribution are shown as follows:

$$q_{\phi_z}(z_i | e_{:,i}, X) = N(\mu_{z_i}, \sigma_{z_i}). \quad (8)$$

We utilize the multi-head self-attention mechanism and the GAT network as the temporal encoder of each sequence, which can capture both complex temporal dependencies and neighbor information. First, we use an LSTM network to encode the relative position information of time series and convert the input value x_i^t at every timestamp t of sequence i to the input embeddings vectors of dimension d . Then we can get the input of multi-head self-attention module h_i^t , where $h_i^t \in R^d$, and $h_i \in R^{w \times d}$. For a sequence of token representations $h_i \in R^{w \times d}$, the self-attention function first projects them into queries $Q \in R^{w \times d_k}$, keys $K \in R^{w \times d_k}$ and values $V \in R^{w \times d_v}$, M times with different linear projections matrixs respectively. Then we apply the scaled dot-product to get the weights

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V. \quad (9)$$

Multi-head attention [38] applies different mappings to learn different representations and output the concatenated feature

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_M)W^O, \quad (10)$$

in which, M is the number of heads. The m th head of the i th sequence is defined as

$$\text{head}_m = \text{Attention}(h_i W_m^Q, h_i W_m^K, h_i W_m^V), \quad (11)$$

where the projections are parameter matrices $W_m^Q \in R^{d \times d_k}$, $W_m^K \in R^{d \times d_k}$, $W_m^V \in R^{d \times d_v}$ and $W^O \in R^{hd_v \times d}$. We note the output of multi-head attention as V_{e_i} , which is the temporal feature representations of sequence i . Then the temporal feature representations of all sequences are sent to the GAT network to aggregate neighbor information according the latent graph relationships learned by the spatial encoder.

$$V'_{e_i} = \text{Relu}\left(\alpha_{i,i} W V_{e_i} + \sum_{j \in N(i)} \alpha_{j,i} W V_{e_j}\right), \quad (12)$$

where $N(i) = \{j | e_{ji} = 1\}$, W is a trainable weight parameter matrix, and $\alpha_{j,i}$ is attention coefficients computed as follow:

$$\xi(i, j) = \text{LeakyRelu}(a^\top (V_{e_i} \oplus V_{e_j})) \quad (13)$$

$$\alpha_{ij} = \frac{\exp(\xi(i, j))}{\sum_{\rho \in N(i) \cup \{i\}} \exp(\xi(i, \rho))}. \quad (14)$$

Finally, the updated temporal feature representations V'_{e_i} is sent to the linear network to get the parameters of Gaussian distribution. For the time window $\{t - w, t - w + 1, \dots, t - 1\}$ of sequence i ,

$$[\mu_{z_i}, \log \sigma_{z_i}] = \text{Linear}(V'_{e_i}). \quad (15)$$

Sampling of latent time series representation: We use the reparameterization trick [13] to obtain the latent time series representation. For the time window $\{t - w, t - w + 1, \dots, t - 1\}$ of sequence i ,

$$z_i = \mu_{z_i} + \sigma_{z_i} \odot \varepsilon, \varepsilon \sim N(0, 1). \quad (16)$$

D. Reconstruction Decoder Network

To model the temporal dependence in latent time series representations and realize the reconstruction of the input sequence, we propose the reconstruction decoder network. We denote $p_{\theta_r}(x_i | z_i, e_{:,i})$ using the reconstruction decoder network, which can construct the historical time series x_i based on the latent time series representation z_i . We utilize a multi-head self-attention network as the reconstruction decoder, whose input is the latent time series representation z_i after linear transformation. The output of the single-head self-attention network is

$$\begin{aligned} & \text{Attn}(z_i W_r^Q, z_i W_r^K, z_i W_r^V) \\ &= \text{Softmax}\left(\frac{z_i W_r^Q \cdot z_i W_r^{KT}}{\sqrt{d_k}}\right) z_i W_r^V, \end{aligned} \quad (17)$$

where the $W_r^Q \in R^{d \times d_k}$, $W_r^K \in R^{d \times d_k}$, $W_r^V \in R^{d \times d_v}$ are linear projection matrices. And *Attn* is an abbreviation for *Attention*. Then we apply multi-head attention following the (10) and (11) to get the concatenated multi-head output:

$$V_{r_i} = \text{MultiHead}(z_i W_r^Q, z_i W_r^K, z_i W_r^V). \quad (18)$$

Finally, the reconstruction output $\{\hat{x}_i^{t-w}, \hat{x}_i^{t-w+1}, \dots, \hat{x}_i^{t-1}\}$ of sequence i can be obtained through the MLP network.

E. Forecasting Decoder Network

To learn more robust latent time series representation and predict the value of future timestamps, we propose the forecasting decoder network. We denote the $p_{\theta_f}(y_i | z_i, e_{:,i})$ using the forecasting decoder network, which can generate the future time series y_i based on the latent representation z_i . The forecasting decoder network is mainly composed of a multi-head self-attention network and an encoder-decoder attention network. The input of the former is a token which is a random initiated vector whose dimension is d and the output $\{y_i^t, y_i^{t+1}, \dots, y_i^{t+w'-2}\}$ at the previous timestamp. They form an input matrix, denoted as $u_i \in R^{w' \times d}$. The output of the single-head attention network is

$$\text{Attn}(u_i W_f^Q, u_i W_f^K, u_i W_f^V)$$

$$= \text{Softmax} \left(\frac{u_i W_f^Q \cdot u_i W_f^{K^T}}{\sqrt{d_k}} \right) u_i W_f^V. \quad (19)$$

Then we apply multi-head attention following the (10) and (11) to get the concatenated multi-head output:

$$V_{f_i} = \text{MultiHead}(u_i W_f^Q, u_i W_f^K, u_i W_f^V), \quad (20)$$

where $V_{f_i} \in R^{w' \times d}$, and $W_f^Q \in R^{d \times d_k}$, $W_f^K \in R^{d \times d_k}$, $W_f^V \in R^{d \times d_v}$ are linear projection matrices. In the encoder-decoder attention network, the queries come from the output of multi-head self-attention decoder network, and the keys and values come from the latent time series representation learned by the encoder, which allows every timestamp in the decoder to attend over all timestamps in the input sequence. The output of the single-head encoder-decoder attention network can be represented as

$$\begin{aligned} & \text{Attn}(V_{f_i} W_{f'}^Q, z_i W_{f'}^K, z_i W_{f'}^V) \\ &= \text{Softmax} \left(\frac{V_{f_i} W_{f'}^Q \cdot z_i W_{f'}^{K^T}}{\sqrt{d_k}} \right) z_i W_{f'}^V. \end{aligned} \quad (21)$$

Then we apply multi-head attention following the (10) and (11) to get the concatenated multi-head output:

$$V'_{f_i} = \text{MultiHead}(V_{f_i} W_{f'}^Q, z_i W_{f'}^K, z_i W_{f'}^V), \quad (22)$$

where $V'_{f_i} \in R^{w' \times d}$, and $W_{f'}^Q \in R^{d \times d_k}$, $W_{f'}^K \in R^{d \times d_k}$, $W_{f'}^V \in R^{d \times d_v}$ are linear projection matrices.

Finally, the prediction output $\{y_i^t, y_i^{t+1}, \dots, y_i^{t+w'-1}\}$ of sequence i can be obtained through the MLP network.

F. Objective Function and Training

According to the theory of VAE [13], we can train our model by optimizing ELBO. The ELBO of the i th sequence has been shown in (3). Considering all sequences, the optimization objective of HSTVAE can be represented as

$$L(X, \phi_e, \phi_z, \theta_r, \theta_f) = \sum_{i=1}^N L(x_i, \phi_e, \phi_z, \theta_r, \theta_f), \quad (23)$$

where we denote the prior Gaussian distribution $p(z_i)$ as $N(0, 1)$ and denote the prior Bernoulli distribution as $Ber(b)$. The b is a constant which controls the sparsity of the learned graph and can be determined by experiment. Moreover, $\phi_e, \phi_z, \theta_r, \theta_f$ means the parameters of the spatial encoder network, temporal encoder network, reconstruction decoder network and forecasting decoder network respectively, and we utilize the stochastic gradient descent (SGD) algorithm to update them during the training process.

G. Anomaly Detection

Based on the fact that anomalies often occur in a continuous period of time, we propose a history-future combined anomaly score mechanism. The reconstruction error at timestamp $t = \{t - w, t - w + 1, \dots, t - 1\}$ of sequence i is $Err_r^i(t) = \sqrt{(\hat{x}_i^t - x_i^t)^2}$ and the forecasting error of sequence i in $t = \{t, t + 1, \dots, t + w' - 1\}$ is denoted as $Err_f^i(t)$. We

TABLE III
STATISTICAL SUMMARY OF THE THREE REAL-WORLD DATASETS

	SWaT	WADI	SMD
Number of Sequences	50	98	28*38
Training size (only normal data)	47519	78457	708405
Testing size (data with anomalies)	44931	17280	708420
Anomaly rate (%)	12.21	5.76	4.16

take the maximum reconstruction error and forecasting error across all sensors at timestamp t as the reconstruction error and forecasting error at timestamp t , and we note them as $Err_r(t)$ and $Err_f(t)$. Since the reconstruction output is $\hat{x}_i^{t-w:t-1}$, we take the maximum reconstruction error in the history window as the historical anomaly factor $A_h(t)$ at current timestamp. The output of multi-step forecasting decoder is $y_i^{t:t+w'-1}$, and we take the maximum prediction error in the future window as the future anomaly factor $A_f(t)$. We weight the historical anomaly factor $A_h(t)$, the future anomaly factor $A_f(t)$, and the forecasting error $A_c(t)$ of current timestamp to get the anomaly score $A_1(t)$ at current timestamp. Moreover, since the sparsity of the graph learned in normal pattern is within a certain range, but each sequence has few related neighbors in attacked scenarios, we take the average neighbor numbers of all sequences as another anomaly score $A_2(t)$ of anomaly detection. Specifically, for the timestamp t , the anomaly score based on reconstruction and forecasting error can be calculated by

$$A_1(t) = A_c(t) + \gamma \cdot A_h(t) + \eta \cdot A_f(t). \quad (24)$$

Where γ and η are the hyper-parameters, which can be determined by grid search [1]. The anomaly score based on graph sparsity can be calculated by

$$A_2(t) = \frac{1}{N} \sum_{i=1}^N D_i, \quad (25)$$

where D_i is the number of neighbors of the i th sequence, N is total number of sequences. We use the grid search to find the optimal threshold thr_1 and thr_2 . When $A_1(t) > thr_1$ or $A_2(t) < thr_2$, we assume an anomaly occurs at timestamp t and the anomaly label $a^t = 1$.

V. EXPERIMENTS

A. Datasets Description

We conduct extensive experiments on three real-world public MTS datasets: SWaT,² WADI,³ SMD.⁴ Table III summarises the statistics of the three datasets.

- **SWaT** The SWaT dataset is collected from a six-stage water treatment testbed in May 2015. The data is collected from 51 sensors and actuators with the 1-second intervals within 11 days, in which the system was in normal operation for the first week, and suffered a total of 41 physical attacks in the following days.

²<https://itrust.sutd.edu.sg/testbeds/secure-water-treatment-SWaT/>

³<https://itrust.sutd.edu.sg/testbeds/water-distribution-wadi/>

⁴<https://github.com/smallcowbaby/OmniAnomaly>

- **WADI** The WADI dataset is collected from a three stages water distribution testbed. The data is produced from 123 sensors and actuators at 1-second intervals within 16 days, of which 14 days are normal data, and the remaining days are attacked 15 times in total.
- **SMD** The Server Machine Dataset contains metrics like CPU load, network usage, memory usage, etc. The data is collected within 5 weeks. And the dataset has 28 different entities, where each entity has 38 metrics.

To improve the training and testing efficiency, the SWaT and WADI data samples are downsampled at 10 s intervals. Specifically, we retain the median of the 10 seconds data and use the label with the highest frequency as the overall label, while SMD uses the original data. In addition, we have normalized all datasets and remove the meaningless sequences that are both 0 on the training set and testing set after normalization. The training set is normal data, while the testing set contains anomalies. We randomly divide 20% of the data in the training set as validation set for model selection. Especially, different entities need to be trained and tested independently in the SMD dataset and we report the mean value.

B. Experimental Setup

Baselines: We compare the performance of our proposed method with thirteen popular anomaly detection methods:

- **PCA [34]:** The principal component analysis uses the reconstruction error of the low dimensional projection as anomaly score.
- **KNN [29]:** The K Nearest Neighbors calculates the average distance between each sample point and its nearest K samples to get the anomaly score.
- **LSTM-VAE [11]:** The LSTM-VAE takes LSTM as the feedforward network of VAE to reconstruct time series.
- **OmniAnomaly [10]:** The OmniAnomaly is a stochastic recurrent neural network which consider the time dependence between latent representations.
- **MAD-GAN [19]:** The MAD-GAN trains LSTM-RNNs on the MTS data and then utilize both the discriminator and the generator to detect anomalies.
- **GRELEN [7]:** The Graph Relational Learning Network combines VAE with a graph dependence structure learning strategy and consider the sum of all the sensors' in/out degree as the anomaly score.
- **GDN [22]:** The Graph Deviation Network uses a top- k graph learning strategy to learn a graph of inter-sequence relationships.
- **GTA [1]:** The GTA uses an improved transformer to learn time dependencies and designs a connection learning policy to learn inter-sequence relationships.
- **DVGCRN [6]:** The Deep Variational Graph Convolutional Recurrent Network combines the proposed EPN with GCRN by a sharing embedding channel and considers both forecasting and reconstruction losses.
- **MTAD-GAT [21]:** The MTAD-GAT is a framework based on two GAT networks, which can separately learn time dependence and inter-sequence relationship of the MTS

data. And the forecasting error and the reconstruction probability are combined to detect anomalies.

- **LSTM-AD [35]:** LSTM-AD uses stacked LSTM networks to learn normal pattern and use predicted errors to evaluate the likelihood of anomalies occurring at each moment.
- **HealthESN [36]:** HealthESN focuses on the issue of data imbalance and designs a new method based on the classical echo state network (ESN) and a threshold selection method based on information theory, which can be more adaptive to different anomaly detection tasks.
- **Telemanom [8]:** Telemanom applies LSTM to anomaly detection and proposes a dynamic thresholding technique that is independent of labels or incorrect parameter assumptions.

Metrics: We take the Precision, Recall, F1 score and ROC-AUC as the evaluation metrics. Moreover, multiple evaluation strategies have been proposed in previous work [10], [39], [40], [41], and we adopt the point-adjust [10] to calculate the metrics. If one point in the anomaly duration is detected, we assume the specific anomalous segments have been detected.

Training Settings: We implement our proposed method using Pytorch version 1.10.0 with CUDA 11.1 and conduct all experiments on four NVIDIA GeForce RTX 3090 GPUs. We set the historical window length to 15 frames to predict the value of next 6 frames and construct the historical 15 frames. Our temporal and spatial encoder network are 1 layer, while the reconstruction and prediction decoder network are 2 layers. The dimension of multi-head attention mechanism in all networks are set to 64 and the number of heads is 8. In the temporal encoder, the dimension of latent variable z is 3. In the spatial encoder, the parameter of Bernoulli prior distribution of SWaT, WADI and SMD datasets are 0.1, 0.2, 0.13 respectively. For our model and all baselines, we use the validation set to help hyper-parameters selection and conduct the grid search [7] on different parameter combination. The model is trained using the Adam optimizer with learning rate 0.001 for up to 50 epochs and early stopping strategy is applied with patience of 5.

C. Performance and Analysis

In Tables IV and V, we study the anomaly detection performance in terms of precision, recall, F1-score and ROC-AUC on our proposed HSTVAE method and other thirteen baselines on datasets SWaT, WADI and SMD. Similar with the GRELEN [7], GTA [1], OmniAnomaly [10], for our HSTVAE and all baselines, we apply a grid search on all possible anomaly thresholds to search for the best F1-score in theory and report them. We run HSTVAE and all baselines with 5 times and report the average and standard deviation. Our proposed HSTVAE significantly outperforms all other methods on three datasets especially in terms of recall, F1 and ROC-AUC. In detail, we observe the following phenomenons from the Tables IV and V:

- The F1 score and recall of HSTVAE outperform all baselines, and the performance of HSTVAE on precision and recall is quite balanced. Especially, we achieve the best F1-score as 0.9007 for SWaT, 0.8431 for WADI and 0.9206 for SMD. Compared to the suboptimal model, HSTVAE

TABLE IV
EXPERIMENTAL RESULTS OF DIFFERENT APPROACHES ON SWAT, WADI AND SMD USING PRECISION, RECALL AND F1 AS METRICS

Datasets Method/Metric	SWaT			WADI			SMD		
	Pre.	Rec.	F1	Pre.	Rec.	F1	Pre.	Rec.	F1
PCA	22.43±0.15	24.12±0.06	23.24±0.14	22.20±0.14	19.01±0.11	20.48±0.12	55.27±0.13	45.77±0.12	50.06±0.12
KNN	9.65±0.09	6.02±0.06	7.40±0.08	9.93±0.15	14.85±0.11	11.91±0.13	71.73±0.08	15.24±0.11	25.15±0.06
LSTM-VAE	95.72±0.36	60.56±1.73	74.16±0.51	75.75±2.39	26.44±1.24	39.02±1.92	72.37±1.29	79.66±1.48	75.28±1.28
OmniAnomaly	77.68±0.56	88.26±0.21	82.64±0.36	39.49±5.22	84.85±3.64	54.53±1.69	88.74±2.11	90.28±1.14	89.04±1.09
MAD-GAN	90.64±0.87	64.25±1.20	75.10±0.91	43.24±0.42	34.61±0.77	38.20±0.61	27.61±1.11	85.48±0.61	41.78±0.56
GRELEN	96.72±1.91	74.13±1.62	83.92±0.37	73.92±1.09	70.26±1.82	72.00±0.88	93.63±0.85	84.00±1.03	88.15±0.85
GDN	94.71±2.12	74.01±2.96	82.98±2.57	70.32±3.89	63.26±2.78	65.86±2.11	87.13±0.56	91.26±1.27	89.06±0.75
GTA	95.20±1.97	83.87±1.52	88.33±0.80	<u>81.45±5.71</u>	79.08±1.51	80.69±3.85	88.64±0.87	<u>86.71±0.69</u>	<u>87.59±0.96</u>
DVGCRN	89.30 ±2.94	83.68±3.46	86.62±2.70	78.38±1.43	83.81±1.08	81.03±1.91	91.57±1.06	86.91±0.72	89.02±0.80
MTAD-GAT	93.66±4.35	74.88±3.19	83.02±0.32	47.79±0.11	<u>86.89±3.93</u>	61.63±1.03	87.93±0.93	88.73±0.84	88.30±0.40
LSTM-AD	95.77±3.31	68.30±2.50	79.60±0.01	90.00±0.69	26.04±0.56	40.31±0.96	87.82±0.66	85.60±0.41	86.40±0.56
HealthESN	99.37±0.01	72.13±0.05	83.59±0.04	73.96±0.06	29.80±0.03	42.48±0.02	86.29±0.05	89.04±0.04	87.70±0.03
Telemanom	84.62±3.56	65.55±1.61	73.48±2.89	50.03±3.27	35.50±2.38	39.43±1.60	63.52±0.22	66.44±0.68	64.70±0.55
HSTVAE(Ours)	89.05 ±0.12	91.15±0.22	90.07±0.19	81.12±0.15	87.89±0.12	84.31±0.13	88.98±0.12	95.39±0.08	92.06±0.11

The best performances are highlighted in bold and the second best are underlined.

TABLE V
EXPERIMENTAL RESULTS OF DIFFERENT APPROACHES ON SWAT, WADI AND SMD USING ROC-AUC AS METRICS

Datasets	SWaT	WADI	SMD
PCA	50.00±0.01	50.01±0.01	70.15±0.18
KNN	24.50±0.04	50.04±0.04	57.53±0.09
LSTM-VAE	60.03±2.12	50.60±2.22	65.67±1.01
OmniAnomaly	74.75±0.24	55.46±0.62	75.32±0.13
MAD-GAN	77.43±0.88	50.90±0.44	51.65±0.33
GRELEN	84.45±0.70	78.55±4.00	86.86±0.50
GDN	83.59±2.38	77.10±1.05	86.45±0.15
GTA	83.83±2.39	<u>79.04±0.88</u>	85.63±0.37
DVGCRN	80.71±0.71	66.93±1.76	74.77±2.76
MTAD-GAT	82.95±0.45	69.84±0.84	81.33±0.54
LSTM-AD	81.82±0.02	51.24±0.45	74.27±0.28
HealthESN	84.38±0.02	50.52±0.02	80.48±0.03
Telemanom	81.86±0.21	56.96±0.25	75.88±0.17
HSTVAE(Ours)	85.11±0.09	80.33±0.14	87.05±0.17

The best performances are highlighted in bold and the second best are underlined.

can achieve 3.27%, 1.15%, 4.52% increase in terms of the recall on three datasets respectively. In addition, our HSTVAE achieved optimal results on ROC-AUC, indicating that our HSTVAE can achieve excellent positive and negative sample differentiation ability without the influence of threshold selection. Specifically, on the SWAT, WADI and SMD datasets, the AUC of HSTVAE is 0.78%, 1.63%, 0.22% higher than that of the optimal baseline, respectively.

- The deep learning-based methods (LSTM-VAE, MAD-GAN, GTA etc.) have better detection performance on three datasets than the conventional approaches such as PCA, KNN. This is because the former includes the recurrent mechanism or attention mechanism which are able to model long-term temporal dependencies. Our HSTVAE uses the attention mechanism for temporal dependencies modeling, which is better than the RNN-based model. Specifically, our F1 is 21.45 %, 116.07%, 22.29% higher than LSTM-VAE on three datasets significantly.
- The methods that consider the graph relationship between sequences, such as GDN, GRELEN, GTA and our HSTVAE can achieve better performance than the methods that only consider temporal dependencies, which shows the importance of modeling relationship among sequences.

Further, compared with the GDN considering only top- k neighbors, the GRELEN, GTA and our HSTVAE retain all potential neighbors, which can avoid the information loss and achieve better performance. The F1 of our HSTVAE achieves 8.66%, 28.01%, 3.37% improvement over GDN on three datasets.

- It's meaningful to model the temporal stochasticity and spatial stochasticity of multivariate time series. The models that do not consider temporal and spatial stochasticity, such as LSTM-AD, HealthESN and Telemanom often achieve high precision, but their recall and F1 are very low. In addition, their AUC also performs worse on the WADI dataset. The OmniAnomaly uses a stochastic recurrent neural network to learn the robust temporal latent representations while the GRELEN employs a latent graph relationship variable to capture the inter-sensors dependence with spatial stochasticity. Both of them have achieved good performance on recall whose values are close to ours, but the precisions of the OmniAnomaly and GRELEN on WADI are 51.32% and 8.88% lower than ours because another stochasticity is not considered.

D. Ablation Studies

To study the effectiveness of each component of our model, we gradually exclude the modules to observe how the model performance degrades on the three datasets. Table VI shows the anomaly detection performance of all variants has declined compared with HSTVAE.

- $HSTVAE_{w/o\ GAT}$: To study the benefit of learned graph, which was learned from the spatial encoder and used to aggregate neighbor information for the time series in temporal encoder by the GAT module, we remove the GAT.
- $HSTVAE_{w/o\ recon.}$: To study the effect of reconstruction decoder, we remove the reconstruction decoder and only perform multi-step prediction.
- $HSTVAE_{w/o\ fore.}$: To study the effect of forecasting decoder, we remove the forecasting decoder and only use the historical reconstruction error to calculate anomaly score.

TABLE VI
EXPERIMENTAL RESULTS OF ABLATION STUDIES ON SWAT, WADI AND SMD USING PRECISION, RECALL, F1 AND ROC-AUC AS METRICS

Datasets	SWaT				WADI				SMD			
Method/Metric	Pre.	Rec.	F1	AUC	Pre.	Rec.	F1	AUC	Pre.	Rec.	F1	AUC
HSTVAE (Ours)	89.05	91.15	90.07	85.11	81.12	87.89	84.31	80.33	88.98	95.39	92.06	87.05
<i>w/o GAT</i>	91.85	84.56	88.05	84.76	77.59	76.13	76.86	76.49	89.27	90.62	89.93	79.29
<i>w/o recon.</i>	83.90	89.84	86.76	84.78	86.60	62.07	72.31	75.75	89.33	89.93	89.62	84.38
<i>w/o fore.</i>	91.12	80.33	85.39	84.87	67.36	83.16	74.43	75.77	87.75	82.57	85.08	70.02
<i>w/o LSTM</i>	88.66	90.46	89.55	84.85	80.52	87.88	84.04	79.24	88.28	91.03	89.63	86.88

The HSTVAE without removal of any components achieves the best performance.

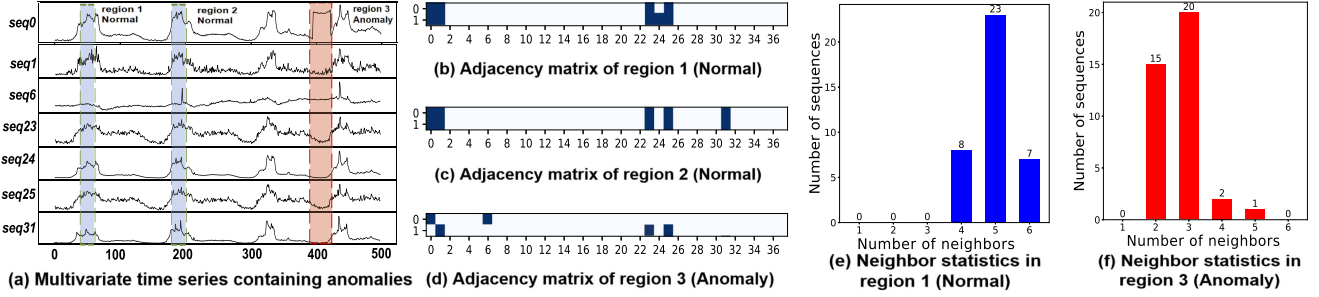


Fig. 4. Visualization of spatial stochasticity. The adjacency matrixs are shown in Fig. 4(b)–(d) and several relevant sequences are shown in Fig. 4(a). In addition, in order to observe the distribution of the number of neighbors of each sequence under normal and anomalous conditions, we make statistics in the Fig. 4(e)–(f).

- $HSTVAE_{w/o LSTM}$: To study the impact of different position encoding methods on model performance, we replace LSTM relative position encoding with sine cosine absolute position encoding.

Table VI shows the performance of the variants, from which we can observe that the anomaly detection performance of all the variants has declined compared with that of HSTVAE. In particular, the recall of $HSTVAE_{w/o GAT}$ decreases significantly, which demonstrates the importance of the graph learning for aggregating proper neighbor information. The recall and AUC of the $HSTVAE_{w/o recon.}$ and $HSTVAE_{w/o fore.}$ both decrease obviously, which proves that the reconstruction decoder and forecasting decoder all play an important role in modeling stochasticity. After replacing LSTM relative position encoding with sine cosine absolute position encoding, the precision, recall, f1 and AUC of $HSTVAE_{w/o LSTM}$ all decreased, indicating the necessity of modeling relative position information.

E. Interpretability

In order to show the spatial stochasticity of our model, we visualize the adjacency matrix of the $seq0$ and the $seq1$ in three special time periods of the SMD machine-1-1 dataset.

Spatial stochasticity: The spatial stochasticity is reflected in the fact that the learned graph relationships between sequences are dynamic with time changing. As shown in the Fig. 4(b)–(d), $seq0, 1, 23, 24, 25$ are similar in region 1. In region 2, $seq0, 1, 23, 25, 31$ are similar. However, in anomaly region 3, there has been a significant change in the graph relationship, where $seq0, 6$ are similar, while $seq1, 23, 25$ are similar.

Normal and anomaly: Comparing Fig. 4(e) and (f), we can see that the average number of neighbors of the sequence in normal mode is in the range of 4–6. However, when the anomaly

occurs, the number of neighbors is concentrated in the range of 2–4, which is consistent with the phenomenon in Fig. 4(a). From Fig. 4(a), in Region 1 and Region 2, there are 4 or 5 sequences that are similar to the $seq0$, while in Region 3, because the $seq0$ fluctuates violently, its pattern is no longer similar to other normal sequences, so its neighbors are reduced. The phenomenon that the sequence has few neighbors when anomalies occur shows that we can use the sparsity of the learned graph to identify the anomaly.

F. Case Study

In order to show the importance of modeling temporal stochasticity and spatial stochasticity in anomaly detection, we visualize the anomaly scores and thresholds of our HSTVAE and MAD-GAN, GRELEN, GDN. We perform case study on the time period shown in Fig. 4(a), which includes temporal fluctuations, relationship fluctuations and anomalies. As shown in Fig. 5, in the normal time period on the left, our HSTVAE achieves the most smoothest score, which reflects its high robustness. The MAD-GAN misjudged near $t = 50$, $t = 200$ and $t = 300$ where fluctuation occurs, which may be because it only considers temporal stochasticity and ignores spatial stochasticity. In addition, GRELEN considers spatial stochasticity and has no misjudgment, but its robustness is not as good as HSTVAE, manifested as the anomaly score of noises being close to the threshold. As a deterministic methods, the GDN mistakenly detected fluctuations as anomalies at $t = 200$, which also shows the necessity of considering temporal and spatial stochasticity. In the anomalous time period marked by the red rectangle, all the methods have detected the anomaly, but our model can more clearly separate anomalies and normal from a global perspective,

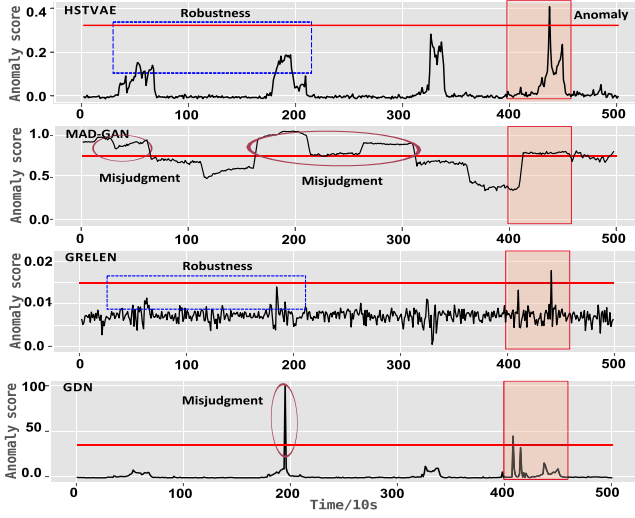


Fig. 5. Anomaly score and threshold of HSTVAE, MAD-GAN, GRELEN, GDN when temporal fluctuation, relationship fluctuation and anomaly occur. The black curve represents anomaly scores, and the red line represents the threshold. The blue rectangular dotted line box indicates the area with fluctuation but normal and correctly detected as normal. The areas of misjudgment are circled with oval red solid line frame and the ground truth anomaly areas are marked by red rectangular shadow.

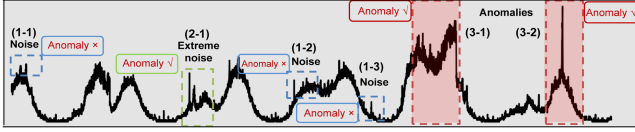


Fig. 6. Noises and Anomalies. The "√" and "X" in the elliptical boxes adjacent to each region represent whether our model detects that region as an anomaly.

which reflects the good anomaly detection performance of our model.

To further explore the anomaly detection performance of our model in the presence of noise [9], [42], [43], especially extreme noise, we visualize some common noises, extreme noises and anomalies from our used real-world dataset SMD in Fig. 6. As shown in the region (3-1) and region (3-2) of Fig. 6, the shapes of the time series deviate from the historical pattern significantly and experience drastic fluctuations, thus being detected as anomalies. From the region (1-1), (1-2), and (1-3) in Fig. 6, our model shows strong robustness against ordinary noises, which would not misjudge the points as anomalies. Finally, facing with the extreme noises in region (2-1), our proposed HSTVAE also classifies them as anomalies. For anomaly detection tasks, it can be considered reasonable because we need to recognize as many anomalies as possible while ensuring performance for better maintenance decisions. This also reflects that our HSTVAE can achieve the best recall comparing with other baselines.

G. Parameter Analysis

To investigate the influence of different hyper parameters that effect the performance of our model, we conduct experiments on following parameter variants. Especially, when analyzing

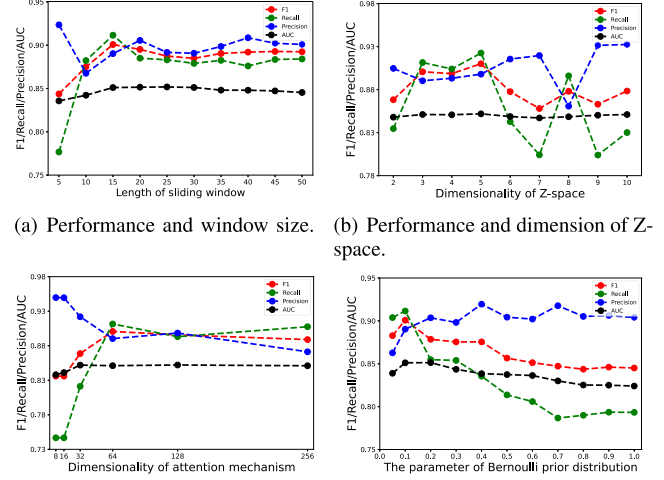


Fig. 7. Hyper-parameter analysis on SWaT dataset.

a specific parameter, we fix the other three parameters as the parameters reported in Section V-B.

- *Window size*: When evaluating the impact of window size on model performance, we fix the dimensionality of latent representation z to 3, the attention mechanism dimension to 64, and the parameter of Bernoulli prior distribution to 0.1. The length of historical window can affect the time dependence modeling performance of the temporal encoder and the graph relationship learning performance of the spatial encoder. Fig. 7(a) shows that the recall, f1 and AUC increase with the increase of window size and reach the peak at 15, and then decrease slightly. The result confirms the long-term time dependence modeling ability of the attention mechanism.
- *The dimensionality of Z-space*: When evaluating the impact of the dimensionality of latent representation z on model performance, we fix the window size to 15, the attention mechanism dimension to 64, and the parameter of Bernoulli prior distribution to 0.1. The size of latent representation z affects the representation ability of the latent time series representation variables. When the value is too small, the features of the input data cannot be learned, and if the value is too large, some redundant information may be retained. Fig. 7(b) shows that the performance first increases and then decreases with the increase of the dimensionality of latent representation z , and reaches the optimal performance when it is 5.
- *The dimension of attention network*: When evaluating the impact of the dimension of attention network on model performance, we fix the window size to 15, the dimensionality of latent representation z to 3, and the parameter of Bernoulli prior distribution to 0.1. Different dimensions of attention network have various effect on the long-term temporal dependency modeling capability of the temporal encoder. Fig. 7(c) shows that the performance of the model becomes better with the increase of dimensions,

TABLE VII
EXPERIMENTAL RESULTS OF HSTVAE ON SWaT, WADI AND SMD USING DIFFERENT EVALUATION STRATEGIES FOR ANOMALY DETECTION

Datasets	SWaT			WADI			SMD		
Method/Metric	Pre.	Rec.	F1	Pre.	Rec.	F1	Pre.	Rec.	F1
HSTVAE+Poi.Aj	89.05	91.15	90.07	81.12	87.89	84.31	88.98	95.39	92.06
Point-wise	99.28	60.44	75.14	73.25	40.40	52.07	51.76	45.17	48.24
Segment-wise	83.57	64.34	72.70	61.54	57.14	59.26	64.68	56.34	60.22
Range-based	84.34	57.02	68.04	60.92	35.15	44.58	50.59	38.39	43.65
Affiliation metrics	86.27	81.65	83.90	76.24	83.02	79.49	89.28	90.30	89.79

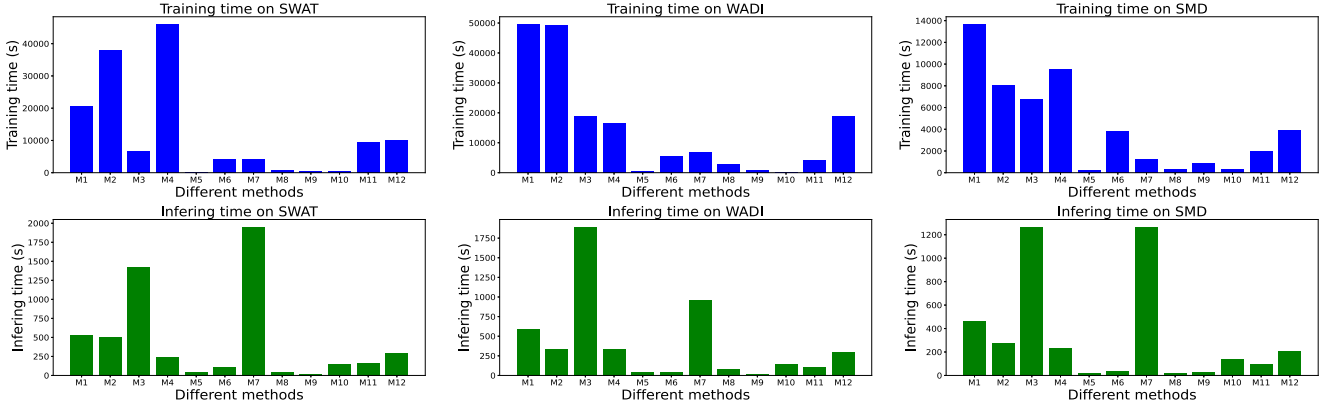


Fig. 8. Training and inference time of different approaches on SWaT, WADI and SMD datasets. The unit is seconds(s). For the convenience of drawing, we use M1-M12 to represent "LSTM-VAE, OmniAnomaly, MAD-GAN, GRELEN, GDN, GTA, DVGCRN, MTAD-GAT, LSTM-AD, HealthESN, Telemanom, HSTVAE" in sequence.

and reaches the optimal value at 64, and then tends to be stable. In addition, AUC only showed slight changes with the increase of the dimension of attention network, indicating the robustness of the model.

- *The parameter of Bernoulli prior distribution:* When evaluating the impact of the parameter of Bernoulli prior distribution on model performance, we fix the window size to 15, the dimensionality of latent representation z to 3, and the dimension of attention network to 64. The Bernoulli prior determines the sparsity of the learned graph. Fig. 7(d) shows that when the prior probability is 0.1, the model achieves the optimal performance. When this value is larger or smaller, the performance will decline, which means that richer latent representations can be learned by combining appropriate neighbor information, while too many neighbors will cause information redundancy and even introduce error information.

H. Evaluation Strategy Analysis

In order to investigate the impact of different evaluation strategies on anomaly detection results, we conduct experiments with several common evaluation strategies [10], [39], [40], [41]. The point-wise and segment-wise methods [39] consider a time point and a continuous anomalous period as an event to calculate metrics, respectively. The range-based method [40] considers the size and relative position of the overlap between the predicted

and actual anomaly set in each event. The affiliation metrics [41] consider the temporal adjacency and events durations.

From Table VII, it can be seen that the point-adjust strategy [10] we adopted is most suitable for the three datasets in this paper, and our model *HSTVAE* also achieves good results when using the affiliation metrics.

I. Efficiency Analysis

To analyze the efficiency of our model, we visualize the training and inference time of HSTVAE and baselines. As shown in Fig. 8, the training and inference time of our HSTVAE is at a moderate level to show the considerable operational efficiency while achieving optimal performance. Overall, the reconstruction based methods such as OmniAnomaly, MAD-GAN and GRELEN often require more training and inference time than prediction based methods such as GDN, GTA, LSTM-AD and HealthESN, which may be due to their complex network structures designed to learn robust latent representations. In addition, models that consider inter sequence relationships and perform feature aggregation, such as GRELEN, may require more training time.

VI. CONCLUSION

In this paper, we proposed an end-to-end hierarchical spatio-temporal variational autoencoder framework, which can address the temporal stochasticity and spatial stochasticity within MTS simultaneously. We modeled the relationship between sequences

as a binary random variable, and automatically learned the directed graph structure through the spatial encoder. And a GAT network was applied to combine the graph with the temporal encoder to embed the highly structured input time series into latent time series representation that can capture both complex temporal dependencies and neighbor information. Then the reconstruction decoder and forecasting decoder constructed the historical sequence and predicted the future sequence according to the latent time series representation and achieved the anomaly detection through a history-future combined anomaly score mechanism. Extensive experiments on three real-world datasets demonstrated that our method can achieve the best and balanced performance compared with other multivariate time series anomaly detection approaches. We also conducted visualization analysis and case studies to show the interpretability of our framework. In the future work, we consider to adopt dynamic threshold selection to make *HSTVAE* more suitable for online real-time anomaly detection scenarios.

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Xiao Zhang is now an assistant professor with the School of Computer Science and Technology, Shandong University. His research interests include data mining, multi-task learning and federated learning. He has published more than 20 papers in the prestigious refereed journals and conference proceedings, such as *IEEE Transactions on Mobile Computing*, *UBICOMP*, *ACM Multimedia*, *IJCAI*, *AAAI*, *ACM CIKM*, and *IEEE ICDM*.



Shuqing Xu is currently working toward the MD degree with the School of Computer Science and Technology, Shandong University. Her research interests include data mining and anomaly detection.



Huashan Chen received the BS and MS degrees from Shandong University, and the Institute of Information Engineering, Chinese Academy of Sciences, respectively, and the PhD degree in computer science from the University of Texas at San Antonio, in 2021. He is currently a Young associate professor with the Institute of Information Engineering, Chinese Academy of Sciences. His research interests include cybersecurity metrics, automated attack planning, and blockchain security.



Zekai Chen received the PhD degree in computer science from George Washington University, in 2021. He is a senior research scientist with Amazon. His research areas include foundational large language model, multi-modal deep learning, multi-task learning, and recommendation. He has published several influential papers in top-tier conferences and journals, including but not limited to *AAAI*, *ACL*, *ICDM*, *CIKM*, *WACV*, *IEEE Internet of Things Journal*, *IEEE Transactions on Mobile Computing*, *IEEE Transactions on Computers* and others.



Fuzhen Zhuang received the BE degree from the College of Computer Science, Chongqing University, Chongqing, China, in 2006, and the PhD degree in computer science from the Institute of Computing Technology, Chinese Academy of Sciences, Beijing, China, in 2011. He is currently a full professor with the Institute of Artificial Intelligence, Beihang University. He has published more than 140 papers in some prestigious refereed journals and conference proceedings such as *Nature Communications*, *IEEE Transactions on Knowledge and Data Engineering*, *IEEE Transactions on Cybernetics*, *IEEE Transactions on Neural Networks and Learning Systems*, *ACM Transactions on Knowledge Discovery from Data*, *ACM Transactions on Intelligent Systems and Technology*, *Information Sciences*, *Neural Networks*, *SIGKDD*, *IJCAI*, *AAAI*, *TheWebConf*, *SIGIR*, *ICDE*, *ACM CIKM*, *ACM WSDM*, *SIAM SDM*, and *IEEE ICDM*. His research interests include transfer learning, machine learning, data mining, multitask learning, knowledge graph, and recommendation systems. He is a senior member of the CCF. He was a recipient of the Distinguished Dissertation Award of CAAI, in 2013.



Hui Xiong (Fellow, IEEE) received the PhD degree in computer science from the University of Minnesota. He is a chair professor, associate vice president (Knowledge Transfer), and the Thrust of Artificial Intelligence, The Hong Kong University of Science and Technology (Guangzhou), China. And he is with the Department of Computer Science and Engineering, The Hong Kong University of Science and Technology Hong Kong SAR, China. His research interests span artificial intelligence, data mining, and mobile computing. He has served on numerous organization and program committees for conferences, including as program co-chair for the Industrial and Government Track for the 18th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD), Program co-chair for the IEEE 2013 International Conference on Data Mining (ICDM), General co-chair for the 2015 IEEE International Conference on Data Mining (ICDM), and Program co-chair of the Research Track for the 2018 ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. He received several awards, such as the 2021 AAAI Best Paper Award and the 2011 IEEE ICDM Best Research Paper award. For his significant contributions to data mining and mobile computing, he was elected as a Fellow of AAAS, in 2020.



Dongxiao Yu received the BS degree from the School of Mathematics, Shandong University, in 2006, and the PhD degree from the Department of Computer Science, The University of Hong Kong, in 2014. He became an associate professor with the School of Computer Science and Technology, Huazhong University of Science and Technology, in 2016. He is currently a professor with the School of Computer Science and Technology, Shandong University. His research interests include edge intelligence, distributed computing and data mining.